Riemann Sums and Definite Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_0^4 x^2\,dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; right endpoints $1, 2, 3, 4$: sum $= 1(1) + 1(4) + 1(9) + 1(16) = 30$.

2 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_1^5 (2x + 1)\,dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; left endpoints $1, 2, 3, 4$: values $3, 5, 7, 9$: sum $= 24$.

3 [Foundational] • Target: The definite integral and its properties

The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2)\,dx$ exactly.

$\left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.

4 [Foundational] • Target: The definite integral and its properties

Evaluate $\int_{-2}^2 (x^3 - 4x)\,dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

The integrand is odd ($f(-x) = -f(x)$), and the interval is symmetric about 0, so the integral is 0.

5 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_0^4 x^2\,dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.

$\Delta x = 1$; midpoints $0.5, 1.5, 2.5, 3.5$: sum $= 0.25 + 2.25 + 6.25 + 12.25 = 21$. Close to the exact value 21.33.

6 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.

$\Delta x = 1$, values at $1, 2, 3, 4, 5$: $1, 0.5, 0.333, 0.25, 0.2$. Trapezoidal:

$\frac{1}{2}(1 + 2(0.5 + 0.333 + 0.25) + 0.2) = \frac{1}{2}(1.2 + 2.166) \approx 1.683$.

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7 [Foundational] • Target: Signed area

The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x-2)\,dx$ and explain why the result does not equal the total area between the curve and the axis.

$\left[\frac{x^2}{2} - 2x\right]_0^4 = (8-8) - 0 = 0$. The integral gives signed (net) area: the region below the axis on $[0, 2]$ cancels the region above it on $[2, 4]$, even though both have positive area.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Signed area

Find the total (unsigned) area between $f(x) = x - 2$ and the x-axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.

$\left|\int_0^2 (x-2)\,dx\right| + \left|\int_2^4 (x-2)\,dx\right|$. $\int_0^2 (x-2)\,dx = \left[\frac{x^2}{2} - 2x\right]_0^2 = (2-4) - 0 = -2$, so $|-2| = 2$. $\int_2^4 (x-2)\,dx = \left[\frac{x^2}{2} - 2x\right]_2^4 = (8-8) - (2-4) = 0 - (-2) = 2$. Total area $= 2 + 2 = 4$.

9 [Intermediate] • Target: Comparing Riemann sum estimates

For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

Underestimate — since f is increasing, the left endpoint of each subinterval gives the smallest function value on that subinterval, so each rectangle sits below the curve.

10 [Intermediate] • Target: Comparing Riemann sum estimates

Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

The right-endpoint estimate (30) overestimates the true value (≈ 21.33), consistent with $f(x) = x^2$ being increasing — right endpoints give the largest function value on each subinterval, so the rectangles sit above the curve.

11 [Intermediate] • Target: Riemann sums from a table of values

A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 9 + 14 + 20 = 48$.

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12 **[Intermediate]** • **Target: The definite integral as a limit of Riemann sums**

Write the formal limit definition of the definite integral $\int_a^b f(x)\,dx$ in terms of a Riemann sum, and explain the role of $n \to \infty$.

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$. As $n \to \infty$, the rectangles become infinitely thin, and the sum converges to the exact signed area under the curve, regardless of which sample point x_i^* (left, right, midpoint) is used within each subinterval.

13 **[Intermediate]** • **Target: Riemann sums with sigma notation**

Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2\,dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.

$R_n = \sum_{i=1}^{n} f(x_i)\Delta x = \sum_{i=1}^{n} \left(1 + \frac{2i}{n}\right)^2 \cdot \frac{2}{n}$.

14 **[Intermediate]** • **Target: Riemann sums with sigma notation**

Estimate $\int_0^6 f(x)\,dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.

$\Delta x = 2$; midpoints are exactly $1, 3, 5$: sum $= 2(4 + 10 + 16) = 60$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Comparing approximation methods on a concave function** • **Cross-Lesson: 13.01 Vectors and Vector-Valued Functions**

For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.

For concave-down f, the trapezoids lie below the curve (underestimate), while midpoint rectangles' tops cross above the curve on average (overestimate). So the midpoint estimate is larger, and the trapezoidal estimate underestimates the true area.

16 **[Challenging]** • **Target: The definite integral as a limit of Riemann sums** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

Compute $r'(\theta) = -\sin\theta$. Form the radicand: $r^2 + (r')^2 = (1+\cos\theta)^2 + (-\sin\theta)^2 = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta = 2(1+\cos\theta)$. Recall the half-angle identity $1 + \cos\theta = 2\cos^2(\theta/2)$: $r^2 + (r')^2 = 2\left(2\cos^2(\theta/2)\right) = 4\cos^2(\theta/2)$. Taking the square root for $\theta \in [0, \pi]$: $\sqrt{4\cos^2(\theta/2)} = 2\cos(\theta/2)$. By symmetry across $[0, 2\pi]$.

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17 **[Challenging]** • **Target: Riemann sums with sigma notation** • **Cross-Lesson: 15.01 Double Integrals**

Level 1 · FoundationsExpress the following limit as a definite integral with lower limit $a = 0$, and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2}{n}\sqrt{4-\left(\frac{2i}{n}\right)^2}$

Identify components: $\Delta x = \frac{2}{n}$, so with $a = 0$, the upper limit is $b = 2$. Sample points are $x_i = \frac{2i}{n}$. The integrand is $f(x) = \sqrt{4-x^2}$. The limit becomes: $\int_0^2 \sqrt{4-x^2}\,dx$ Geometrically, $y = \sqrt{4-x^2}$ represents the upper half of a circle of radius $r = 2$ centered at the origin. The interval $[0, 2]$ represents one quarter of the circle: $\int_0^2 \sqrt{4-x^2}\,dx = \frac{1}{4}\pi r^2 = \frac{1}{4}\pi(2)^2 = \boldsymbol{\pi} \approx 3.1416$.

18 **[Challenging]** • **Target: Comparing approximation methods on a concave function** • **Cross-Lesson: 16.01 Vector Fields and Line Integrals**

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

Construct the Taylor polynomial using $a = 2$: $P_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$ $= 3 - 1(x-2) + \frac{4}{2}(x-2)^2 + \frac{-12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 - 2(x-2)^3$ Substitute $x = 2.1$, so $x - 2 = 0.1$: $P_3(2.1) = 3 - 0.1 + 2(0.1)^2 - 2(0.1)^3 = 3 - 0.1 + 2(0.01) - 2(0.001)$ $= 3 - 0.1 + 0.02 - 0.002 = 2.918$

19 **[Challenging]** • **Target: Approximating area with Riemann sums** • **Cross-Lesson: 2.01 Evaluating Limits**

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

The field is conservative with potential function $f(x, y, z) = x^3y + 2z$. By the Fundamental Theorem for Line Integrals: $\int_C ablaf \cdot d\mathbf{r} = f(2, 1, 3) - f(0, 0, 1)$ Evaluate at endpoints: $f(2, 1, 3) = 2^3(1) + 2(3) = 8 + 6 = 14$ $f(0, 0, 1) = 0^3(0) + 2(1) = 2$ Therefore: $\int_C \mathbf{F}\cdot d\mathbf{r} = 14 - 2 = 12$

20 **[Challenging]** • **Target: The definite integral and its properties** • **Cross-Lesson: 2.02 Continuity**

Level 1 · FoundationsEvaluate the improper integral $\int_1^\infty \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

Rewrite as a limit: $\lim_{b\to\infty} \int_1^b x^{-3}\,dx = \lim_{b\to\infty} \left[-\frac{1}{2x^2}\right]_1^b = \lim_{b\to\infty} \left(-\frac{1}{2b^2} - \left(-\frac{1}{2(1)^2}\right)\right)$. Since $\lim_{b\to\infty} \frac{1}{2b^2} = 0$: $0 - \left(-\frac{1}{2}\right) = \boldsymbol{\frac{1}{2}}$. The integral converges to $\frac{1}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Approximating area with Riemann sums

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:

a. $n \to \infty$ (partition width $\to 0$) ✓

b. $b \to \infty$

c. $f \to 0$

d. $x \to a$

The definite integral is defined as the limit of Riemann sums as the number of subintervals $n \to \infty$.

22 [Foundational] • Target: The definite integral and its properties

The midpoint rule approximation for $\int_0^1 \int_0^1 f(x, y)\, dA$ with one rectangle:

a. $f(1, 1)$

b. $f(0, 0)$

c. $f(1/2, 1/2)$ ✓

d. $f(0, 1)$

Midpoint rule: sample at midpoint $(1/2, 1/2)$.

23 [Foundational] • Target: Midpoint and trapezoidal approximations

Integral test: $\sum_{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)dx$ converges, for f positive decreasing. So $\sum 1/n^2$ converges because $\int_1^{\infty} x^{-2}dx =$

a. 1 **(finite)** ✓

b. ∞

c. 0

d. 1/2

$\int_1^{\infty} x^{-2}dx = 1$: finite.

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24 [Foundational] • Target: Signed area

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?

a. 4.5

b. 6

c. **7.5** ✓

d. 9

The area of the shaded region is given by the definite integral $\int_1^4 (-x + 5)\,dx$. To evaluate this integral, we find the antiderivative of $f(x) = -x + 5$, which is $F(x) = -\frac{x^2}{2} + 5x$. Now we apply the Fundamental Theorem of Calculus: $\int_1^4 (-x + 5)\,dx = F(4) - F(1)$ $F(4) = -\frac{4^2}{2} + 5(4) = -\frac{16}{2} + 20 = -8 + 20 = 12$ $F(1) = -\frac{1^2}{2} + 5(1) = -\frac{1}{2} + 5 = 4.5$ Therefore, the area is $12 - 4.5 = 7.5$. Common errors include incorrectly calculating the anti.

25 [Foundational] • Target: Comparing Riemann sum estimates

Evaluate the improper integral $\int_0^\infty \frac{1}{x^2 + 9}\,dx$.

a. $\frac{\pi}{3}$

b. ∞

c. $\frac{\pi}{6} + 1$

d. $\frac{\pi}{6}$ ✓

Write it as a limit, e.g. $\lim_{b \to \infty} \int_0^b \frac{1}{x^2 + 9}\,dx$, integrate, and take the limit. The integral converges to $\frac{\pi}{6}$.

26 [Foundational] • Target: Riemann sums from a table of values

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s?

a. 8 m²/s

b. **16 m²/s** ✓

c. 32 m²/s

d. 4 m²/s

Applying Applications of Derivatives principles, $dA/dt = 2s \cdot ds/dt = 2(4)(2) = 16$. Therefore, the verified correct option is 16 m²/s.

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27 [Foundational] • Target: The definite integral as a limit of Riemann sums

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2+y^2} =$

a. always 1

b. **an integer ✓**

c. always 0

d. always π

Applying Applications of Integration principles, Winding number is an integer. Therefore, the verified correct option is an integer.

28 [Foundational] • Target: Riemann sums with sigma notation

Triple integral $\iiint_V 1\,dV$ gives:

a. **volume of V ✓**

b. 0

c. surface area

d. 1

Applying Multiple Integrals principles, Triple integral of 1 = volume. Therefore, the verified correct option is volume of V.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing approximation methods on a concave function

Riemann sum step width $\Delta x =$

a. $1/n$

b. $(b-a)/n$ ✓

c. $n(b-a)$

d. $(f(b)-f(a))/n$

Applying Fundamental Theorem principles, $(b-a)/n$ for n subintervals. Therefore, the verified correct option is $(b-a)/n$.

30 [Intermediate] • Target: Approximating area with Riemann sums

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$ ✓

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

Applying Applications of Integration principles, $I_x = bh^3/12$. Therefore, the verified correct option is $bh^3/12$.

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31 [Intermediate] • Target: The definite integral and its properties

A rectangle has perimeter 40. Find the width w that maximises its area.

a. **10** ✓

b. 20

c. 5

d. $\frac{40}{3}$

With length $\ell = \frac{40}{2} - w$, $A(w) = w(\frac{40}{2} - w)$; $A'(w) = \frac{40}{2} - 2w = 0 \Rightarrow w = \frac{40}{4} = 10$.

32 [Intermediate] • Target: Midpoint and trapezoidal approximations

$\int_0^1 \int_0^x y\,dy\,dx$: inner integral gives $x^2/2$, so outer integral $=$

a. 1

b. 1/4

c. 1/2

d. 1/6 ✓

Applying Multiple Integrals principles, $\int_0^1 x^2/2\,dx = 1/6$. Therefore, the verified correct option is $1/6$.

33 [Intermediate] • Target: Signed area

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region S in the uv-plane with area 2 maps to a region R. Find the area of R.

a. 2

b. 8

c. 12

d. 16 ✓

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |8|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 16$.

34 [Intermediate] • Target: Comparing Riemann sum estimates

Green's theorem area: $\text{Area} = \iint_D dA = \frac{1}{2}\oint_C (x\,dy - y\,dx)$. For unit circle:

a. π ✓

b. 2π

c. 1

d. $\pi/2$

Applying Integrals principles, Unit circle area $= \pi$. Therefore, the verified correct option is π.

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35 [Intermediate] • Target: Riemann sums from a table of values

A deltoid (3-cusp hypocycloid): $x = \frac{2}{3}a\cos t + \frac{a}{3}\cos 2t$, $y = \frac{2}{3}a\sin t - \frac{a}{3}\sin 2t$. Its area:

a. $\pi a^2/4$

b. $\pi a^2/3$ ✓

c. $2\pi a^2/3$

d. πa^2

Deltoid area $= \frac{2}{9} \cdot \pi a^2 \cdot (3/2)$... actually the answer is $\pi a^2/3$. I'll keep this.

36 [Intermediate] • Target: The definite integral as a limit of Riemann sums

Green's theorem area formula: area $= \frac{1}{2}\oint_C (x\,dy - y\,dx)$. For ellipse $x = a\cos t$, $y = b\sin t$:

a. $\pi(a + b)$

b. $2\pi ab$

c. πab ✓

d. ab

$\frac{1}{2}\int_0^{2\pi} (a\cos t \cdot b\cos t - b\sin t \cdot (-a\sin t))dt = \frac{ab}{2} \cdot 2\pi = \pi ab$.

37 [Intermediate] • Target: Riemann sums with sigma notation

Generating function of Fibonacci: $F(x) = \sum_{n=0}^{\infty} F_n x^n = \frac{x}{1 - x - x^2}$. The coefficient of x^6:

a. 13

b. 8 ✓

c. 5

d. 21

Applying Sequences and Series principles, $F_6 = 8$. Therefore, the verified correct option is 8.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 16.01 Vector Fields and Line Integrals

For $|x| < 1$, which function equals $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$?

a. $\frac{1}{1-x}$

b. $\frac{1}{(x+1)^2}$

c. $\ln(x + 1)$

d. $-\ln(1 - x)$ ✓

Integrate the geometric series 1/(1–x) term by term: the sum is $-\ln(1 - x)$.

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39 [Challenging] • Target: Approximating area with Riemann sums • Cross-Lesson: 2.01 Evaluating Limits

A solid cylinder $x^2 + y^2 \le 16,\ 0 \le z \le 4$ has density $\delta = \sqrt{x^2 + y^2}$. Find its mass.

a. 64π

b. $\frac{1024\pi}{3}$

c. $\frac{256\pi}{3}$

d. $\frac{512\pi}{3}$ ✓

$m = \int_0^{2\pi}\int_0^4\int_0^4 r \cdot r\,dr\,dz\,d\theta = \frac{512\pi}{3}$.

40 [Challenging] • Target: The definite integral and its properties • Cross-Lesson: 2.02 Continuity

Laurent series: a function analytic in an annulus $r < |z| < R$ has a series $\sum_{n=-\infty}^{\infty} a_n z^n$. For $1/(z(1-z))$ near $z = 0$:

a. $1/z^2 + 1/z + \cdots$

b. $1/z - 1 + z - \cdots$

c. $z + z^2 + \cdots$

d. $1/z + 1 + z + z^2 + \cdots$ ✓

$1/(z(1-z)) = 1/z \cdot \frac{1}{1-z} = \frac{1}{z}(1 + z + z^2 + \cdots) = 1/z + 1 + z + \cdots$

41 [Challenging] • Target: Midpoint and trapezoidal approximations • Cross-Lesson: 3.01 The Derivative from First Principles

A lamina occupies $[0, 3] \times [0, 3]$ with density $\rho = xy$. Find its mass.

a. $\frac{81}{2}$

b. $\frac{81}{4}$ ✓

c. $\frac{81}{8}$

d. 9

$m = \int_0^3\int_0^3 xy\,dx\,dy\ =\ \frac{81}{4}$.

42 [Challenging] • Target: Signed area • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Area of disk $x^2 + y^2 \le R^2$ by polar coordinates:

a. πR

b. $2\pi R$

c. πR^2 ✓

d. $4\pi R^2$

$\int_0^{2\pi}\int_0^R r\,dr\,d\theta = 2\pi \cdot R^2/2 = \pi R^2$.

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43 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

$\int_0^2 \int_0^{2-x} dy\,dx$: region is a triangle. Area $=$

a. 1

b. 4

c. 2 ✓

d. 3

Triangle with vertices $(0,0), (2,0), (0,2)$: area $= 2$.

44 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Apply Fubini: $\int_0^\infty \int_0^\infty e^{-xy}\sin x\,dy\,dx$. Inner integral: $\int_0^\infty e^{-xy}dy =$

a. x

b. e^{-x}

c. $1/x$ ✓

d. 1

$\int_0^\infty e^{-xy}dy = [-e^{-xy}/x]_0^\infty = 1/x$.

45 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 6.01 Related Rates

The Euler-Mascheroni constant $\gamma = \lim_{n\to\infty}(H_n - \ln n) \approx 0.5772$ where $H_n = \sum_{k=1}^{n} 1/k$. Its rationality or irrationality:

a. known rational

b. known irrational

c. zero

d. **unknown (open problem)** ✓

γ unknown rational/irrational — open problem.

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